Chapter 7B

Hemolytic Anemias, Sickle Cell Disease, and RBC Membrane/Enzyme Disorders

Part VII. Hematology and Oncology

Chapter goal
Build a practical illness-script framework for anemia caused by premature red blood cell destruction. The central task is to prove hemolysis, localize intravascular versus extravascular destruction, separate immune from nonimmune processes, recognize life-threatening microangiopathic hemolysis, and connect mechanism to treatment. This chapter is original Medvale synthesis using the uploaded anemia/hemolysis source as a coverage signal, not as copied text.

Why this matters clinically

Hemolytic anemia is not a single diagnosis; it is a physiologic pattern. A patient with fatigue, jaundice, dark urine, gallstones, splenomegaly, back pain, thrombosis, or acute kidney injury may all be showing the same underlying event: red cells are being destroyed faster than the marrow can replace them. The marrow can often increase erythropoiesis several-fold, so hemolysis may be partially compensated until infection, folate depletion, renal disease, pregnancy, or marrow suppression unmasks it. A hemolytic process becomes dangerous when it is rapid, intravascular, immune-mediated, or microangiopathic because these forms can produce shock, hyperkalemia, hemoglobinuric kidney injury, stroke, pulmonary hypertension, or multiorgan failure.

The uploaded hematology source begins the hemolytic anemia framework by emphasizing premature RBC destruction, a normal or reactive marrow response, reticulocytosis, and classification by intrinsic versus extrinsic RBC defects and by intravascular versus extravascular location. This chapter expands that framework into a diagnostic and management segment suitable for graph-RAG extraction.

Core illness scripts

Pattern	Typical clue cluster	Mechanistic anchor	Immediate concern
Warm autoimmune hemolytic anemia	Subacute anemia, jaundice, splenomegaly, positive direct antiglobulin test for IgG; may follow autoimmune disease, lymphoid malignancy, infection, or drugs.	IgG-coated RBCs are removed mainly by splenic macrophages.	Rapid Hb fall; need steroids, transfusion support if unstable, and evaluation for secondary cause.
Cold agglutinin disease	Acrocyanosis, hemolysis worse with cold exposure, positive complement on DAT, possible Mycoplasma/EBV or clonal B-cell disease association.	IgM fixes complement at cooler temperatures; C3-coated RBCs are cleared in liver or lyse intravascularly.	Avoid cold exposure; warm blood/products; rituximab-based therapy if severe.
G6PD deficiency	Episodic jaundice/dark urine after oxidant drugs, fava beans, or infection; bite cells and Heinz bodies.	Low NADPH reduces glutathione regeneration, making RBCs vulnerable to oxidative membrane injury.	Stop trigger; supportive care; test may be falsely normal during acute crisis.
Hereditary spherocytosis	Family history, chronic hemolysis, splenomegaly, pigment gallstones, high MCHC, spherocytes, negative DAT.	Membrane cytoskeleton defect causes surface area loss and splenic trapping.	Folate support; splenectomy in selected severe cases; vaccinate before splenectomy.
Microangiopathic hemolytic anemia	Anemia + thrombocytopenia + schistocytes; neurologic, renal, pregnancy, sepsis, malignant HTN, or drug context.	RBCs shear across platelet/fibrin-rich microthrombi or damaged vasculature.	TTP is a medical emergency: plasma exchange should not wait for ADAMTS13 if suspicion is high.
Sickle cell disease	Vaso-occlusive pain, hemolytic anemia, infection risk, acute chest syndrome, stroke, priapism, avascular necrosis.	HbS polymerizes when deoxygenated, causing sickling, hemolysis, endothelial activation, and vaso-occlusion.	Pain control, infection assessment, oxygenation, hydration without overload, transfusion/exchange when indicated.

Pathophysiology: the hemolysis equation

Normal RBC lifespan is about 120 days. Hemolysis becomes clinically important when RBC destruction accelerates enough that marrow compensation fails or creates measurable reticulocytosis. The laboratory signature reflects three linked processes: cell breakdown releases hemoglobin and LDH, heme catabolism increases indirect bilirubin, and marrow compensation increases reticulocytes if iron, folate, B12, erythropoietin, and marrow architecture are adequate. A patient with hemolysis but no reticulocytosis has a second problem: aplastic crisis, marrow infiltration, severe nutrient deficiency, renal failure, or inflammation blunting erythropoietin response.

> High-yield mechanism
> Hemolysis answers two questions: Are RBCs being destroyed? and why are they being destroyed? Reticulocytes, LDH, bilirubin, haptoglobin, and smear prove the pattern; DAT, smear morphology, family history, medication/infection exposure, thrombosis, and platelet count identify the cause.

Intravascular vs extravascular hemolysis

Extravascular hemolysis occurs when macrophages in the spleen and liver remove abnormal or antibody-coated RBCs. It often produces jaundice, splenomegaly, pigment gallstones, and reticulocytosis. Intravascular hemolysis occurs when RBCs rupture within the circulation. It is more likely to cause hemoglobinemia, hemoglobinuria, hemosiderinuria, acute kidney injury, severe back/flank pain, and very low haptoglobin. Many diseases have mixed features; the labels describe the dominant site, not an absolute location.

Feature	Predominantly extravascular	Predominantly intravascular
Major site	Spleen/liver macrophages	Bloodstream, complement-mediated lysis, mechanical destruction
Classic diseases	Warm AIHA, hereditary spherocytosis, sickle cell disease, thalassemia	PNH, G6PD crisis, cold complement hemolysis, transfusion reaction, MAHA, mechanical valve
Labs	Indirect bilirubin up, LDH up, retics up; haptoglobin may fall but can be less dramatic	LDH often very high, haptoglobin very low/undetectable, plasma free hemoglobin, urine hemoglobin/hemosiderin
Exam/history	Splenomegaly, jaundice, pigment gallstones	Dark urine, back pain, AKI, thrombosis, shock in severe cases

Diagnostic approach

The first step is confirming a hemolytic physiology rather than assuming all jaundiced anemia is hemolysis. Order CBC with differential, reticulocyte count/index, peripheral smear, LDH, total and indirect bilirubin, haptoglobin, direct antiglobulin test, and urinalysis. Add CMP/creatinine, PT/PTT/fibrinogen/D-dimer if DIC or MAHA is possible, type and screen if unstable, and pregnancy testing when relevant. The smear should be reviewed early because schistocytes, spherocytes, bite cells, target cells, sickled cells, and agglutination each point to different disease mechanisms.

Original diagnostic algorithm

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ANEMIA + RETICULOCYTOSIS OR JAUNDICE/DARK URINE
-> confirm hemolysis: LDH up, indirect bilirubin up, haptoglobin down, retics up
-> review smear and platelet count
Schistocytes + thrombocytopenia -> TMA/DIC/HELLP/malignant HTN/prosthetic valve
Spherocytes -> DAT positive = warm AIHA; DAT negative = hereditary spherocytosis
Bite/blister cells -> oxidant injury, especially G6PD deficiency
Sickled cells/target cells -> hemoglobinopathy; confirm with electrophoresis/HPLC
Agglutination -> cold agglutinin disease; warm specimen and check DAT C3
-> DAT result
Positive IgG -> warm AIHA; search secondary causes
Positive C3 only -> cold agglutinin or complement-mediated process
Negative -> membrane, enzyme, hemoglobin, mechanical, PNH, or toxin cause
-> identify emergencies: TTP, DIC, acute chest syndrome, severe transfusion reaction, hyperkalemia, AKI
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Hemolysis labs and their traps

Test	Expected in hemolysis	Pitfall
Reticulocyte count/index	Usually increased; index >2 suggests marrow response	May be low in aplastic crisis, folate/B12 deficiency, marrow disease, renal failure, or early hemolysis.

Test	Expected in hemolysis	Pitfall
LDH	Increased from RBC breakdown and tissue injury	Very high LDH is not specific; also rises with liver injury, ischemia, malignancy, and sample hemolysis.
Indirect bilirubin	Increased from heme catabolism	Severe liver disease can blur direct vs indirect patterns.
Haptoglobin	Low because it binds free hemoglobin	Haptoglobin is an acute-phase reactant and may be normal in inflammation despite hemolysis.
DAT/Coombs	Positive in immune hemolysis	A negative DAT does not exclude all immune hemolysis; a positive DAT can occur without clinically important hemolysis.
Urine heme dipstick	Can be positive with few/no RBCs in intravascular hemolysis	Also positive with myoglobin; correlate with microscopy and CK.

Immune hemolytic anemia

Autoimmune hemolytic anemia is caused by antibodies against the patient's own RBCs. It can be primary or secondary to systemic lupus erythematosus, chronic lymphocytic leukemia, non-Hodgkin lymphoma, immunodeficiency, viral infection, or medications. The direct antiglobulin test identifies immunoglobulin and/or complement bound to the RBC surface. Treatment depends on whether the antibody behaves as a warm IgG process or a cold complement-mediated process.

Feature	Warm AIHA	Cold agglutinin disease
Antibody biology	Usually IgG, active at body temperature	Usually IgM with complement fixation at cold peripheral temperatures
DAT pattern	IgG positive, sometimes C3 positive	C3 positive, IgG usually negative
Hemolysis site	Mostly splenic extravascular	Liver-mediated C3 clearance plus possible intravascular complement lysis
Clues	Spherocytes, jaundice, splenomegaly, other autoimmune disease or lymphoid malignancy	Acrocyanosis, livedo, symptoms worse in cold, RBC agglutination, Mycoplasma/EBV or clonal B-cell disease
Initial management	Glucocorticoids if clinically significant; transfuse if unstable using best-matched blood	Avoid cold, warm patient and blood products; rituximab-based therapy in severe/persistent disease

Transfusion in AIHA
Do not withhold transfusion from an unstable patient solely because compatibility testing is difficult. Blood bank coordination is essential. The practical goal is the safest available unit, not a perfect crossmatch that never arrives.

Drug-induced immune hemolysis

Drugs can cause immune hemolysis by acting as haptens on RBC membranes, creating immune complexes, or inducing true autoantibodies. Classic associations include penicillin-type hapten reactions, cephalosporins, methyldopa-like autoantibody mechanisms, and drug-dependent complement activation. The management principle is immediate discontinuation of the offending drug, supportive transfusion if needed, and immunosuppression when a warm-AIHA-like pattern persists.

RBC membrane disorders

Membrane disorders reduce RBC deformability. Because splenic cords are narrow, poorly deformable RBCs become trapped and removed. This creates chronic extravascular hemolysis with splenomegaly, jaundice, pigment gallstones, and reticulocytosis. The classic high-yield entity is hereditary spherocytosis, but the broader concept is that shape determines splenic survival.

Disorder	Core defect	Smear/labs	Clinical links	Management
Hereditary spherocytosis	Spectrin/ankyrin/band 3/protein 4.2 defects -> membrane loss	Spherocytes, high MCHC, retics up, DAT negative; abnormal osmotic fragility or EMA binding	Autosomal dominant common; jaundice, splenomegaly, gallstones; aplastic crisis with parvovirus B19	Folate; cholecystectomy if stones; splenectomy for selected severe disease after vaccination
Hereditary elliptocytosis	Spectrin self-association or membrane skeletal defects	Elliptocytes; variable hemolysis	Often mild/asymptomatic; severe forms overlap with hereditary pyropoikilocytosis	Usually supportive; folate if chronic hemolysis
Acanthocytosis/spur cells	Altered membrane lipid composition	Spiculated RBCs	Advanced liver disease, abetalipoproteinemia, neuroacanthocytosis	Treat underlying disease

RBC enzyme disorders

Mature RBCs cannot synthesize new proteins and depend on glycolysis for ATP and the pentose phosphate pathway for protection from oxidative injury. Enzyme defects often produce episodic hemolysis triggered by physiologic stress. The key diagnostic trap is timing: testing during or immediately after a crisis may miss deficient older RBCs because the circulating cells are younger reticulocytes with higher enzyme activity.

Disorder	Mechanism	Triggers/clues	Smear	Management
G6PD deficiency	Impaired NADPH generation -> reduced glutathione defense -> oxidative membrane damage	Infection, fava beans, dapsone, primaquine, rasburicase, sulfa drugs, nitrofurantoin; X-linked; episodic dark urine/jaundice	Bite cells, blister cells; Heinz bodies on supravital stain	Stop trigger; supportive care; transfuse if severe; repeat enzyme testing after recovery if initial test is normal but suspicion remains
Pyruvate kinase deficiency	Low ATP generation -> rigid RBC membrane and chronic hemolysis	Congenital nonspherocytic hemolysis; neonatal jaundice or chronic anemia	Echinocytes may be seen	Folate, transfusion support, splenectomy in selected cases; disease-specific therapy may be considered by hematology

Paroxysmal nocturnal hemoglobinuria

PNH is an acquired clonal hematopoietic stem cell disorder caused by loss of GPI-anchored complement regulatory proteins, especially CD55 and CD59. The result is complement-mediated intravascular hemolysis, bone marrow failure overlap, and a striking tendency toward venous thrombosis in unusual sites such as hepatic, portal, mesenteric, cerebral, or dermal veins. The historical name overemphasizes nighttime symptoms; hemolysis is complement-driven and can occur at any time.

PNH diagnostic anchors	Clinical meaning
Hemoglobinuria with few/no RBCs on microscopy	Suggests free hemoglobin rather than true hematuria.
Thrombosis at unusual sites	Major cause of morbidity/mortality; think hepatic vein thrombosis/Budd-Chiari.
Pancytopenia or aplastic anemia association	PNH may arise in a marrow failure context.
Flow cytometry for CD55/CD59 or FLAER	Modern diagnostic test; replaces older Ham/sucrose tests.
Complement inhibition	Eculizumab/ravulizumab and related agents reduce complement-mediated hemolysis; vaccination and infection precautions are essential.

Microangiopathic hemolytic anemia and thrombotic microangiopathy

Microangiopathic hemolytic anemia (MAHA) is a smear diagnosis: schistocytes indicate RBC fragmentation in small vessels or high-shear circuits. The critical pattern is hemolytic anemia plus thrombocytopenia, because platelet consumption implies a thrombotic microangiopathy or DIC rather than isolated mechanical hemolysis. These diagnoses must be triaged quickly because TTP can be fatal without urgent plasma exchange.

Entity	Core mechanism	Clinical/lab pattern	Immediate management logic
TTP	Severe ADAMTS13 deficiency -> ultra-large vWF multimers -> platelet microthrombi	MAHA + thrombocytopenia; neurologic changes, renal injury, fever may occur but full pentad is not required; PT/PTT often normal	Send ADAMTS13 but do not wait if suspicion high; start plasma exchange + glucocorticoids; caplacizumab/rituximab per hematology
HUS	Often Shiga toxin endothelial injury; complement-mediated forms also exist	MAHA + thrombocytopenia + kidney-predominant injury, often after bloody diarrhea in children	Supportive care; avoid unnecessary antibiotics/antimotility in suspected Shiga toxin; complement inhibition for atypical HUS
DIC	Systemic coagulation activation with fibrin deposition and factor/platelet consumption	Bleeding and thrombosis; prolonged PT/PTT, low fibrinogen, high D-dimer, schistocytes possible	Treat trigger; blood products guided by bleeding/procedures; manage sepsis, trauma, malignancy, obstetric cause
HELLP	Pregnancy-associated endothelial/liver injury	Hemolysis, elevated liver enzymes, low platelets; RUQ pain, HTN/proteinuria may occur	Obstetric emergency; delivery timing and maternal stabilization
Mechanical hemolysis	Valve/circuit shear	Schistocytes, high LDH, hemoglobinuria; platelet count may be less profoundly low unless circuit consumes platelets	Evaluate prosthetic valve, LVAD, dialysis circuit, severe aortic stenosis

TTP rule

A patient with hemolytic anemia, schistocytes, and thrombocytopenia has TTP until proven otherwise if no better explanation is obvious. Severe neurologic signs, renal injury, and fever support the diagnosis, but waiting for all five classic features delays care.

Sickle cell disease

Sickle cell disease is both a hemolytic anemia and a vaso-occlusive vasculopathy. A single amino-acid substitution creates hemoglobin S, which polymerizes when deoxygenated. Polymerization deforms RBCs, shortens survival, activates endothelium and leukocytes, reduces nitric oxide bioavailability through free hemoglobin scavenging, and produces intermittent vascular obstruction. The clinical phenotype is therefore not explained by anemia alone: pain crises, stroke, acute chest syndrome, splenic dysfunction, infection, kidney disease, pulmonary hypertension, avascular necrosis, leg ulcers, and priapism all reflect vascular injury plus hemolysis.

Sickle cell genotypes and baseline patterns

Genotype	Typical severity	Common notes
HbSS	Severe	Classic sickle cell anemia; chronic hemolysis, vaso-occlusive complications, functional asplenia.
HbS-beta0 thalassemia	Severe	No normal beta chain production; resembles HbSS.
HbSC	Moderate but variable	Less anemia but more viscosity; retinopathy and avascular necrosis are important.
HbS-beta+ thalassemia	Variable/milder	Some HbA production; symptoms range from mild to clinically significant.
Sickle trait	Usually asymptomatic	Can matter with extreme dehydration/hypoxia, renal papillary necrosis, hematuria, exertional collapse risk contexts.

Vaso-occlusive pain episode

Pain episodes are usually precipitated by infection, dehydration, cold exposure, hypoxia, stress, menstruation, high altitude, or no identifiable trigger. Evaluation should not assume pain is uncomplicated: fever, hypoxia, chest pain, focal neurologic deficits, priapism, pregnancy, severe anemia, or renal failure are red flags. Treatment prioritizes rapid analgesia, reassessment, hydration matched to volume status, oxygen only if hypoxemic, and search for infection or acute chest syndrome when clinically suggested. Overhydration can worsen pulmonary complications.

Acute chest syndrome

Acute chest syndrome is a new pulmonary infiltrate with symptoms such as fever, chest pain, cough, tachypnea, wheezing, or hypoxemia in a patient with sickle cell disease. It can be triggered by infection, fat embolism from marrow infarction, hypoventilation from pain/opioids, or pulmonary vaso-occlusion. Management commonly includes oxygen for hypoxemia, incentive spirometry, antibiotics covering typical and atypical organisms, adequate analgesia without respiratory suppression, transfusion when oxygenation worsens or anemia is severe, and exchange transfusion for severe hypoxemia, multilobar disease, or rapid deterioration.

Sickle cell complications by organ system

Organ/system	Complications	Clinical anchor
Spleen/immune	Functional asplenia, encapsulated bacterial infection, splenic sequestration in children	Fever is high risk; vaccination and prophylaxis are prevention pillars.
Brain	Ischemic stroke, silent infarcts, cognitive effects	Transcranial Doppler screening in children; transfusion programs for high-risk patients.
Lung/heart	Acute chest syndrome, pulmonary hypertension	Hypoxemia, chest pain, dyspnea, or new infiltrate changes management.
Kidney	Hyposthenuria, hematuria, papillary necrosis, albuminuria, CKD	Nocturia and inability to concentrate urine can appear early.
Bone	Dactylitis, avascular necrosis, osteomyelitis	Salmonella is a classic board association, but Staphylococcus remains common clinically.
GU	Priapism	Persistent episode is an emergency to prevent erectile dysfunction.
Eye	Proliferative retinopathy	Particularly important in HbSC; needs ophthalmology screening.

Disease-modifying and preventive therapy

Hydroxyurea increases fetal hemoglobin and reduces polymerization, vaso-occlusive episodes, and acute chest syndrome frequency for many patients. Chronic transfusion can prevent recurrent stroke but creates iron overload and alloimmunization risk. Curative approaches include hematopoietic stem cell transplantation in selected patients, and gene-based therapies are emerging in specialized settings. Preventive care is as important as acute care: immunization, penicillin prophylaxis in young children, fever plans, transcranial Doppler screening in children, renal albuminuria screening, ophthalmology screening, and reproductive counseling all reduce long-term morbidity.

Acute sickle emergency	Do not miss	Common first moves
Fever	Sepsis from encapsulated organisms	Cultures, empiric antibiotics, evaluate for acute chest/osteomyelitis/UTI.
Chest pain or hypoxemia	Acute chest syndrome	CXR, oxygen if hypoxemic, antibiotics, incentive spirometry, transfusion assessment.
Neurologic deficit	Stroke	Urgent imaging and exchange transfusion pathway with hematology/neurology.
Severe LUQ pain and shock/anemia	Splenic sequestration	Volume support and urgent transfusion; mainly pediatric.
Persistent priapism	Ischemic injury	Urology emergency; analgesia, aspiration/phenylephrine pathways.

Other hemoglobinopathies and hemolytic patterns

Thalassemias were introduced in the prior anemia segment as microcytic disorders of globin-chain production. In practice, thalassemia overlaps with hemolysis because ineffective erythropoiesis and peripheral RBC destruction both contribute. Hemoglobin C disease, unstable hemoglobins, and mixed hemoglobinopathy states can also present with target cells, chronic hemolysis, gallstones, splenomegaly, or episodic worsening under stress. Hemoglobin electrophoresis or high-performance liquid chromatography clarifies most hemoglobinopathy patterns, but iron deficiency can confound HbA2 interpretation in beta-thalassemia screening.

> Exam clue
> Microcytosis with a normal or high RBC count should trigger thalassemia thinking; microcytosis with high RDW, low ferritin, and low transferrin saturation should trigger iron deficiency thinking. Both can coexist.

Hemolysis and transfusion reactions

Transfusion reactions are an important acquired cause of hemolysis. Acute hemolytic transfusion reactions are usually ABO-related and can present during transfusion with fever, chills, flank pain, hypotension, dyspnea, hemoglobinuria, DIC, and acute kidney injury. Delayed hemolytic reactions occur days to weeks later due to an anamnestic antibody response against non-ABO antigens; patients may have falling hemoglobin, jaundice, and a newly positive antibody screen. The immediate response to suspected acute hemolysis is to stop the transfusion, maintain IV access with normal saline, notify the blood bank, check clerical matching, send blood/urine studies, and treat shock/DIC/AKI.

Reaction	Timing	Mechanism	Clues	Management
Acute hemolytic	Minutes-hours	ABO incompatibility or severe complement hemolysis	Fever, flank pain, hypotension, hemoglobinuria, DIC/AKI	Stop transfusion; blood bank workup; fluids; support BP/renal perfusion
Delayed hemolytic	3-14 days, sometimes later	Memory antibody to RBC antigen	Falling Hb, jaundice, positive DAT/antibody screen	Supportive; antigen-negative blood if needed
Febrile nonhemolytic	During/soon after	Cytokines/leukocyte antibodies	Fever/chills without hemolysis	Stop and evaluate; antipyretics if hemolysis excluded
TRALI/TACO	Within hours	Lung injury vs volume overload	Hypoxemia; pulmonary edema; TACO has volume signs	Respiratory support; diuresis for TACO

Initial management by severity

Management of hemolysis depends on tempo, mechanism, and organ threat. Stable chronic hemolysis may require folate support, vaccination planning, gallstone evaluation, trigger avoidance, and outpatient hematology. Acute severe hemolysis requires stabilization, transfusion planning, electrolyte/renal monitoring, and mechanism-specific treatment. Never let the diagnostic workup delay treatment for TTP, acute chest syndrome, septic fever in sickle cell disease, or an acute hemolytic transfusion reaction.

Clinical situation	Key actions
Unstable anemia or symptomatic oxygen delivery failure	Type and screen/crossmatch, transfuse PRBCs when benefits exceed risks, monitor potassium/calcium/volume, coordinate with blood bank if antibodies present.
Suspected immune hemolysis	DAT with IgG/C3, smear, secondary-cause workup; glucocorticoids for warm AIHA if significant; avoid cold/warm products in cold disease.
Suspected TTP	Do not wait for ADAMTS13 if high suspicion; urgent plasma exchange and hematology consultation.
G6PD crisis	Stop oxidant trigger, treat infection, avoid repeat triggers, transfuse if severe, repeat enzyme test later if needed.
Sickle vaso-occlusive episode	Rapid analgesia, reassessment, evaluate fever/chest/neuro/priapism red flags, hydration by volume status, oxygen only if hypoxemic.
Acute chest syndrome	Antibiotics, oxygen if needed, incentive spirometry, analgesia, transfusion/exchange decision based on severity.

Clinical situation	Key actions
PNH	Flow cytometry diagnosis; evaluate thrombosis/marrow failure; complement inhibition and vaccination planning through hematology.

Differentials and mimics

Not every elevated bilirubin in anemia is hemolysis, and not every positive urine heme dipstick is hemoglobinuria. Hepatitis, biliary obstruction, Gilbert syndrome, rhabdomyolysis, occult bleeding with reticulocytosis, recent transfusion, resorbing hematoma, and sample hemolysis can mimic pieces of a hemolytic picture. Conversely, early hemolysis can be missed if reticulocytes have not risen, if haptoglobin is elevated as an acute phase reactant, or if the patient has marrow disease that prevents compensation.

Mimic	How it confuses	Separator
Gilbert syndrome	Indirect bilirubin elevation	Normal Hb, LDH, haptoglobin, retics; episodic with fasting/illness.
Rhabdomyolysis	Positive urine heme with few RBCs	CK markedly elevated; myoglobin rather than hemoglobin.
Occult bleeding	Reticulocytosis and anemia	Iron deficiency over time; no LDH/haptoglobin hemolysis pattern.
Liver disease	Jaundice, spur cells possible	Direct bilirubin, AST/ALT/alk phos pattern, coagulopathy, imaging.
Sample hemolysis	Spurious LDH/potassium changes	Repeat properly drawn sample; no clinical hemolysis pattern.

High-yield exam clues

- Spherocytes with positive DAT suggest warm AIHA; spherocytes with negative DAT and family history suggest hereditary spherocytosis.
- Schistocytes with thrombocytopenia are a hematologic emergency until TTP, DIC, HUS, HELLP, malignant hypertension, and mechanical causes are sorted.
- Bite cells after infection or oxidant exposure suggest G6PD deficiency; the enzyme assay may be falsely normal during the acute episode.
- Hemoglobinuria, thrombosis in unusual veins, and pancytopenia should trigger PNH testing by flow cytometry.
- Pain crisis in sickle cell disease should not distract from fever, hypoxemia, chest infiltrate, neurologic deficits, or priapism.
- Aplastic crisis in chronic hemolysis presents with a sudden Hb drop and low reticulocyte count, classically from parvovirus B19.
- Acute chest syndrome is not just pneumonia; it is a sickle-cell pulmonary emergency with infection, infarction, fat embolism, and hypoventilation contributors.
- Cold agglutinin disease can artifactually raise MCV and create RBC agglutination; warming the sample may correct the analyzer pattern.

Common pitfalls

- Calling hemolysis from LDH alone. LDH is sensitive but nonspecific; use a pattern.
- Assuming DAT positivity equals clinically important AIHA. Some patients have DAT positivity without active hemolysis.
- Waiting for ADAMTS13 before treating a high-probability TTP presentation.
- Overhydrating sickle cell patients with pain, which can worsen pulmonary edema or acute chest physiology.
- Forgetting folate demand in chronic hemolysis and missing superimposed megaloblastic change.
- Treating all sickle cell pain as uncomplicated when fever, hypoxemia, chest pain, or neurologic symptoms are present.
- Missing delayed hemolytic transfusion reaction when Hb falls days after transfusion.
- Assuming normal haptoglobin excludes hemolysis in inflammatory states.

Practice questions

Question 1: A 24-year-old man develops dark urine and jaundice two days after treatment of pneumonia. Smear shows bite cells. LDH is high and haptoglobin is low. What is the mechanism?
Answer: Oxidative RBC injury due to impaired glutathione defense, classically G6PD deficiency. Infection itself and oxidant drugs can precipitate episodes. Stop triggers and provide supportive care; repeat enzyme testing after recovery if needed.

Question 2: A woman with lupus has Hb 6.8 g/dL, indirect bilirubin elevation, reticulocytosis, spherocytes, and DAT positive for IgG. What is the most likely hemolysis site and first-line disease-specific therapy?
Answer: Splenic extravascular hemolysis from warm autoimmune hemolytic anemia. Glucocorticoids are typical first-line therapy when clinically significant, with transfusion support if unstable.

Question 3: A patient has anemia, platelets 18,000/microL, creatinine 1.7 mg/dL, confusion, LDH 1400 U/L, and schistocytes. PT/PTT are normal. What should happen while ADAMTS13 is pending?
Answer: Treat presumptive TTP urgently with plasma exchange and immunosuppression in coordination with hematology. ADAMTS13 testing helps confirm but should not delay treatment when suspicion is high.

Question 4: A patient with sickle cell disease presents with pain, fever, oxygen saturation 88%, and a new lower-lobe infiltrate. What syndrome is present and what management domains matter?
Answer: Acute chest syndrome. Treat with oxygen for hypoxemia, antibiotics, incentive spirometry, adequate analgesia, and transfusion/exchange transfusion depending on severity.

Question 5: A patient has chronic Coombs-negative hemolysis, high MCHC, splenomegaly, and pigment gallstones. What diagnosis is likely and why does splenectomy help?
Answer: Hereditary spherocytosis. Splenectomy reduces macrophage-mediated removal of poorly deformable spherocytes, improving hemolysis, but it requires careful vaccination and infection-risk planning.

Question 6: A patient has hemoglobinuria, abdominal vein thrombosis, leukopenia, and thrombocytopenia. What diagnostic test should be ordered?
Answer: Flow cytometry for PNH clone detection using GPI-anchored protein markers such as CD55/CD59 or FLAER.

Graph-RAG extraction map

The following map is designed for concept-node and edge extraction. Each concept should be stored with synonyms, diagnostic cues, mechanisms, source references, and relationships to labs, smear findings, complications, and treatments.

Node	Class memberships	Key properties	High-value edges
Hemolytic anemia	Disease pattern; anemia mechanism	Premature RBC destruction; reticulocytosis if marrow intact; LDH/indirect bilirubin up; haptoglobin down	causes -> jaundice; causes -> pigment gallstones; evaluated by -> hemolysis labs; subclass -> intravascular/extravascular
Direct antiglobulin test	Diagnostic test; immunohematology	Detects IgG and/or complement on RBC surface	positive in -> AIHA; differentiates -> warm vs cold immune hemolysis
Schistocytes	Smear finding	Fragmented RBCs from shear injury	suggests -> MAHA; linked to -> TTP/HUS/DIC/mechanical valve
Warm AIHA	Immune hemolysis	IgG-mediated; spherocytes; DAT IgG; splenic clearance	treated by -> glucocorticoids; associated with -> SLE/CLL/lymphoma/drugs
Cold agglutinin disease	Immune hemolysis; complement disease	IgM/C3; cold-induced symptoms; agglutination	managed by -> warming/rituximab-based therapy; associated with -> Mycoplasma/EBV/clonal B cells
G6PD deficiency	Enzyme defect; hemolytic anemia	Oxidant sensitivity; bite cells; X-linked	triggered by -> infection/fava/oxidant drugs; diagnostic trap -> false normal acute assay
Hereditary spherocytosis	Membrane disorder	Spherocytes; high MCHC; DAT negative; splenomegaly	causes -> pigment gallstones; treated by -> folate/splenectomy selected cases
PNH	Complement-mediated hemolysis; marrow failure overlap	CD55/CD59 loss; hemoglobinuria; unusual thrombosis	diagnosed by -> flow cytometry; treated by -> complement inhibition
TTP	TMA; hematologic emergency	ADAMTS13 deficiency; MAHA + thrombocytopenia; neuro/renal findings	treated by -> plasma exchange; do-not-delay -> ADAMTS13
Sickle cell disease	Hemoglobinopathy; hemolytic anemia; vaso-occlusive disease	HbS polymerization; pain crises; acute chest; stroke; infection risk	treated by -> hydroxyurea/transfusion/supportive acute care; causes -> functional asplenia

Suggested edge ontology

- mechanism_of: G6PD deficiency -> oxidative hemolysis; HbS polymerization -> sickling and vaso-occlusion.
- diagnosed_by: warm AIHA -> DAT IgG; PNH -> flow cytometry; sickle cell disease -> hemoglobin electrophoresis/HPLC.
- smear_finding_of: schistocytes -> MAHA; spherocytes -> warm AIHA/hereditary spherocytosis; bite cells -> G6PD deficiency.
- complication_of: pigment gallstones -> chronic hemolysis; acute chest syndrome -> sickle cell disease; thrombosis -> PNH.
- emergency_management_for: TTP -> plasma exchange; acute hemolytic transfusion reaction -> stop transfusion; acute chest syndrome -> oxygen/antibiotics/transfusion evaluation.
- mimics_or_overlaps_with: myoglobinuria <-> hemoglobinuria; iron deficiency <-> thalassemia; DIC <-> TTP in MAHA presentations.

Source and guideline cross-check note

Coverage was guided by the uploaded hematology/anemia source segment, especially its opening hemolytic anemia classification. Clinical accuracy was cross-checked against major hematology guideline resources, including American Society of Hematology sickle cell disease guideline materials, British Society for Haematology autoimmune hemolytic anemia guidance, and ISTH thrombotic thrombocytopenic purpura guidance. This chapter is not a substitute for local protocols or specialist consultation in hematologic emergencies.